

Retroarch3DS~RomMArch Front-End Intergration

All due credit to the developers of RetroArch and RomM. I take no credit for their respective masterworks.

This is a modified front-end of the latest Retroarch3DS Build, to intergrate synchronization with your RomM Database.

It is -only- a front-end replacement of RetroArch3DS. Retroarch3DS still installs cores independently of the front-end,

within which the game plays and retains a separate configuration from the front-end.

Nothing changes with how your cores interact with this front-end.

A new sub-menu is now available only within the front-end, allowing synchronization with your RomM server.

The Synchronization function is manual, NOT automatic:

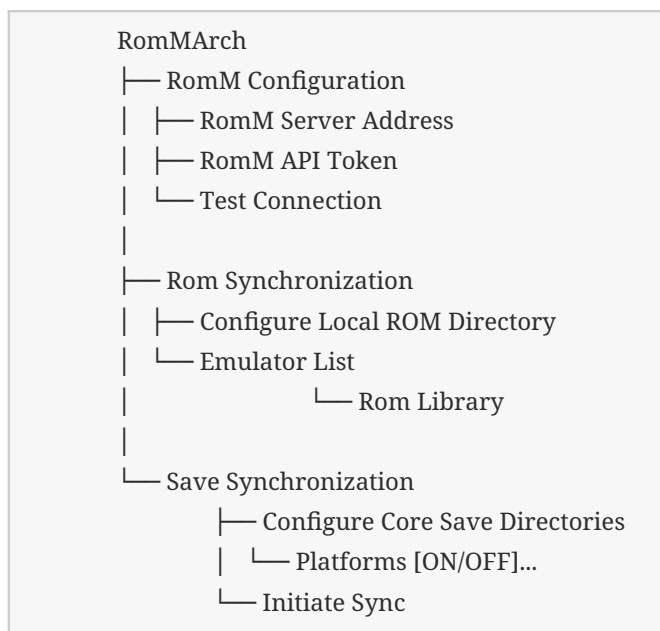
Meaning: if I save a game on my 3DS and wish to push the new save to the RomM server:

I MUST exit the game, re-open the retroarch front-end, navigate and execute the synchronization feature.

Likewise, to if my local save is outdated, the synchronization feature must be executed before the new save is downlaoded from RomM.

Use of this front-end assumes you have a working knowledge of: homebrew 3DS with FBI installer, and the retroarch client.

Menu Flow



Preface

"RomMArch" uses its own configuration file located at /retroarch/rommarch.cfg, alongside your retroarch.cfg. RomMArch creates and maintains this file as configuration is entered.

After continued use, additional information will be written to rommarch.cfg, such as save hashes, which are critical to accurate save synchronization.

It should start out with something like:

```
server_url=http://192.168.1.195:#####
```

```
api_token=rmm_123454321
```

```
roms_path=sdmc:/retroarch/downloads
```

Roms and Saves synchronize separately. Roms can and do download from RomM without synchronizing the RomM save. The reverse is untrue: saves cannot synchronize without their respective rom being downloaded/stored on the 3DS first.

Rom Synchronization means the rom is on both your RomM server and your 3DS.

Save Synchronization reconciles save files for ROMs which are both on your 3DS AND in RomM. The ROM filenames must match exactly between the two locations.

ex: Chrono Trigger.smc and Chrono Trigger (USA).smc will not be recognized as the same ROM, nor synchronize cleanly.

They are represented in two different menu's.

Once you have a game + save synchroinzed, you only have to hit "Initiate Sync" from the "Save Synchronization" menu, to push/pull the save from your RomM database.

SAVE SYNCHRONIZATION CAN ONLY BE UTILIZED AFTER you've used your 3DS Emulator Core to launch at least one game.

The savefile pathway is generated the first time a game is launched. Saving the game is not required for this to happen.

The default savefile path is: sdmc:/retroarch/cores/savefiles/core_name/savefile.srm

If you want to use this tool with an retroarch on your 3DS, but the emulator core hasn't established a savefile directory yet, GO DO THAT FIRST.

RomM Configuration

Server Address

Enter the RomM server base address. RomMArch accepts the server root and handles the API path internally. The value is persisted in rommarch.cfg.

API Token

Enter the RomM Client API token. The token is stored in rommarch.cfg and displayed verbatim in the configuration UI. The current project assumes a token with sufficient RomM access for ROM and save/device synchronization operations.

Test Connection

Test Connection performs an authenticated request to verify that the configured server and token work from the 3DS. This has been confirmed on real hardware.

Rom Synchronization

Rom Library is gated by its prerequisites. Before the RomM emulator/rom browser is exposed, both a server address and API token must be setup in the former step, and a Local ROM Directory must be configured here.

The chosen directory is thereafter permanently stored in the rommarch.cfg file. Erasure of the value/config will gate the library access until reconfigured.

Local ROM Directory

default - roms_path=sdmc:/retroarch/downloads

This is the location where roms are stored on your 3DS. The directory can either be manually entered in your cfg file, or assigned through the interface. Erasure of the .cfg entry will gate access to your RomM Library until reconfigured.

Emulator List

Displays all emulator categories available on your RomM Server

Rom Library

Downloading/Synchronizing

After selecting an emulator, all available roms will be available for download or deletion to/from your local storage.

Roms are already considered synchronized if they already exist in both your local directory and your RomM Library, indicated with a [X] in the rom menu.

Unselecting an existing rom will prompt you to confirm deletion from local storage, with a double-check redundancy.

Deleting

Deleting a locally stored rom has no impact on your RomM Library. It only removes the local rom file from your configured directory. It also does NOT remove any save files that are already created locally.

To download a rom from your RomM Library, in the rom menu, select one/multiple, then choose Synchronize at the bottom of the page.

The game will download to the directory you configured. If you select a rom to download, but return to the Emulator directory without choosing Synchronize,

then the selection will be cleared and it will not be downloaded.

It's faster to manually copy large files directly to your SD card directory, due to the limited processing speed.

If manually copying roms to your SD card rom library, be mindful of consistent file naming between RomM and 3DS storage, to ensure accurate synchronization.

For roms with multiple discs for which you wish to use an M3U configuration, see my note "Regarding M3U's" at the bottom.

Save Synchronization

Save FILES are synchronized.

Save states are explicitly NOT synchronized.

It was an intentional project boundary from the beginning, not an unfinished feature.

RomMArch synchronizes normal in-game save files only. RetroArch save states are not synchronized, modified, or deleted.

If your RomM save has a timestamp suffix, commonly something like~ Chrono Trigger [timestamp information].sav, expect the synchronization function to rename the save

locally on your 3DS, so it's appropriately recognized by the emulator, which expects it to match the rom name, while only changing extension.

ex: A time-stamped on RomM, "Chrono Trigger [timestamp information].sav", would be sync'd down to the 3DS as ~> "Chrono Trigger.sav"

RomMArch uses RomM's server-side update information to determine the most recently updated save. The most recent uploaded save to RomM is the newest save.

SAVE AUTHORITY IS DETERMINED BY HOW RECENT A SAVE WAS UPLOADED TO RomM

Extra words inside the file name do NOT determine save authority

example 1)

Chrono Trigger (USA) [08262055].smc ~ will not be seen as the newest save because it has the later date typed into the savefile name on RomM. (sometime in the year 2055 apparently)

example 2)

Lets say Chrono Trigger (USA).smc was saved to RomM a few hours ago. I did some clerical work and made myself a backup copy called Chrono Trigger (USA)_previous.smc, then placed

it into RomM. Chrono Cross (USA)_previous.smc now becomes the canonical save file because it carries newer server information.

The 3DS clock is NOT used to determine save authority:

Rooted 3DS's date/time are commonly unconfigured, and that's fine.

The synchronization system is resilient to a wrong 3DS date/time.

Save sync only processes ROMs actually present on the 3DS.

Configure Core Save Directories

This tells RomM where to look for the selected console's save file.

NOTE: The core MUST be used to launch a game at least once. This GENERATES THE SAVEPATH, generally located in: `sdmc:/retroarch/cores/savefiles/core_name/savefile.srm`

You'll be required to point RomM to your emulator's save-path.

Due to various emulator cores producing different savepaths, and depending on the user, two people may have different save paths for the same console.

A list of your RomM Library Consoles will be displayed. You'll see the [OFF] status next to each by default. Complete configuration for synchronization will be indicated by [ON].

Choose which console saves you wish to synchronize. In that console's sub-menu, you'll see:

☐ Enable Save Sync

☐ Local Save Path: unconfigured

Complete the setup by enabling and setting the path:

☒ Enable Save Sync

☒ Local Save Path: `sdmc:/retroarch/cores/savefiles/<core-folder>/` && select "Choose this folder"

Initiate Sync

With:

- RomM server and api configured
- a rom synchronized on both RomM and the 3DS
- your save sync enabled & pathway setting

.. RomMArch will scrub local and romM saves against each other and provide you with the results: # Pushed, # Pulled, # Remains the Same, # Conflict Detected

#Pushed = uploaded to RomM

#Pulled = downladed from RomM

#Remains the same = RomM and 3DS files match and nothing is transferred

#Conflict Detected = both the 3DS save and the selected RomM save have changed independently since their last known synchronized state, and their current contents differ.

You'll be given the choices:

choose the local save

choose the RomM save

do nothing

Notes

Regarding M3U's

This is **not** tested with m3u structure for multi-disc games. At the time of this build, I've experienced m3u save-file compatibility errors cross-platform, between the RomM interface and other devices. Therefore, my RomM database has all multi-disc games stored traditionally (Chrono Cross (Disc 1).bin) as separate rom/save directories, and they're stored likewise on my 3DS in the retroarch/downloads directory, without any m3u configuration. They speak to each other in this similar manner. I cannot guarantee that if you have M3U setup on your RomM and 3DS side, that they'll work together the same.

Traditional:

~decades of success

rom_name=Chrono Cross (Disc 1).bin

save_name=Chrono Cross (Disc 1).save_extension

M3U:

~ depending on your setup, a text file and/or folder named Chrono Cross.m3u, are configured with the individual discs, still named:

- Chrono Cross (Disc 1).bin
- Chrono Cross (Disc 2).bin

however, resulting in..

rom_name=Chrono Cross.m3u (regardless of disc)

save_name=Chrono Cross.save_extension (regardless of disc)

Without going much more in depth, again, I'm assuming users have a working knowledge regarding use of their own emulator to perform disc-swap functions, because the RomM interface doesn't play nice with it yet.

With all that said.. expect RomMArch to synchronize the save associated with the individual RomM ROM/disc entry.

Known Errors

HTTP-1

Closing the 3DS lid while the RomMArch frontend is running puts the console to sleep and interrupts its network connection. After reopening the lid, attempting to use RomMArch without restarting the app can produce the HTTP-1 error.

To recover, fully close and reopen the RomMArch frontend after the console wakes, then retry the operation. Reopening the lid alone does not restore the connection used by RomMArch.

To avoid this issue, close the frontend before putting the console to sleep, or restart it after waking before using RomMArch.